

Answer Key & Marking Scheme

Question 1

Part (a): Convert $(64)_7$ to base 9

[3 Marks]

Solution

Step 1: Convert from base 7 to base 10

[1 M]

$$(64)_7 = 6 \times 7^1 + 4 \times 7^0 = 42 + 4 = (46)_{10}$$

Step 2: Convert from base 10 to base 9

[1 M]

Repeatedly divide by 9, recording each remainder:

$$46 \div 9 = 5 \text{ R } 1 \quad \leftarrow \text{LSD (first remainder)}$$

$$5 \div 9 = 0 \text{ R } 5 \quad \leftarrow \text{MSD (last remainder)}$$

Step 3: Read remainders from bottom to top (MSD to LSD)

[1 M]

$$(46)_{10} = (51)_9$$

Final Answer: $(64)_7 = (51)_9$

Part (b): Subtract 45_6 from 54_6 using 6's complement

[3 Marks]

Solution

Step 1: Find the 6's complement of 45_6

5's complement (subtract each digit from 5):

[0.5 M]

$$55_6 - 45_6 = 10_6$$

6's complement (add 1 to the 5's complement):

[0.5 M]

$$(10)_6 + 1 = (11)_6$$

Step 2: Add 54_6 and the 6's complement of 45_6

[1.5 M]

$$\begin{array}{r} 54_6 \\ + 11_6 \\ \hline 105_6 \end{array}$$

Working (right to left):

- Units: $4 + 1 = 5$ (no carry)
- Sixes: $5 + 1 = 6 = 1 \times 6 + 0$, write 0, carry 1
- Carry: 1

Result: 105_6

Step 3: Discard the carry to get the final result

[0.5 M]

Since there is a carry out of the most significant digit, discard it:

$$\cancel{1}05_6 \rightarrow (05)_6 = (5)_6$$

Final Answer: $54_6 - 45_6 = (5)_6$

Question 2.

Part (a)

$$F(A, B, C) = AB + C'$$

w.k.t $A + A' = 1$

— 0.5 marks

$$\Rightarrow F(A, B, C) = AB(1) + C'(1)(1)$$

$$F = AB(C + C') + C'(A + A')(B + B')$$

— 0.5 marks

$$F = ABC + ABC' + (AC' + A'C')(B + B')$$

$$= ABC + ABC' + ABC' + AB'C' + A'BC' + A'B'C'$$

— 0.75 marks

Using $A + A' = 1$

— 0.25 marks

$$F = ABC + ABC' + AB'C' + A'BC' + A'B'C'$$

— 0.5 marks

$$F = \Sigma m(0, 2, 4, 6, 7) \quad \text{— 0.5 marks}$$

Alternate approach

A	B	C	AB	C'	$F = AB + C'$	Minterm
0	0	0	0	1	1	m_0
0	0	1	0	0	0	m_1
0	1	0	0	1	1	m_2
0	1	1	0	0	0	m_3
1	0	0	0	1	1	m_4
1	0	1	0	0	0	m_5
1	1	0	1	1	1	m_6
1	1	1	1	0	1	m_7

So $F = 1$ for:

$$0, 2, 4, 6, 7$$

Therefore:

$$F = \Sigma m(0, 2, 4, 6, 7)$$

— 2 marks for correct truth table

— 1 mark for final answer

Part (b)

$$F(A, B, C, D) = \Sigma(1, 3, 7, 11, 15) + d(5, 13)$$

$$\Rightarrow F = \Pi(0, 2, 4, 6, 8, 9, 10, 12, 14) + d(5, 13) \quad \text{— } 0.5 \text{ marks}$$

K-Map:

$AB \backslash CD$	00	01	11	10
00	0	1	1	0
01	0	×	1	0
11	0	×	1	0
10	0	0	1	0

— **0.75 marks** for correct placement and grouping of 0s and don't cares
0.75 marks for correctly identifying the expression from each group:

- Group of 8 (red) $\rightarrow D'$
- Group of 4 (blue) $\rightarrow AC'$

$$F' = D' + AC'$$

Taking complement on both sides,

$$(F')' = (D' + AC')'$$

$$F = D(A' + C) \quad \text{— } 1 \text{ marks}$$

Question 3

$$A \oplus B = AB' + BA' \quad \text{— } 0.5 \text{ marks}$$

$$[A \oplus B]' = AB + B'A' \quad \text{— } 0.5 \text{ marks}$$

To prove: $(A \oplus B \oplus C) = [(A \oplus B)' \oplus C]'$

LHS:

$$\begin{aligned}
 A \oplus B \oplus C &= [(AB' + BA') \oplus C] \\
 &= [AB' + BA'] \cdot C' + [AB' + BA']' \cdot C \\
 &= AB'C' + BA'C' + A'B'C + BA \cdot C
 \end{aligned} \quad \text{— } 1 \text{ mark}$$

RHS:

$$\begin{aligned} [(A \oplus B)' \oplus C]' &= [(AB + A'B') \oplus C]' \\ &= [(AB + A'B') \cdot C + (AB + A'B')' \cdot C']' \\ &= ABC + A'B' \cdot C + A'B \cdot C' + AB' \cdot C' \end{aligned} \quad \text{--- 1 mark}$$

Hence:

$$A \oplus B \oplus C = [(A \oplus B)' \oplus C]'$$

Total: 3 marks